**Computational Physics I 3630/5630          Fall 2026**

**Homework Assignment 2**

**1. Monte Carlo Integration.** Suppose the volume $V_d$ of the $d$-dimensional unit ball is known. In this problem, use a $d$-dimensional Monte Carlo integral to estimate the volume $V_{d+1}$ of the unit ball in $(d+1)$ dimensions. By slicing the $(d+1)$-dimensional ball at $\mathbf{x} = (x_1, \ldots, x_d)$,

$$V_{d+1} = 2 \int_{|\mathbf{x}| \leq 1} \sqrt{1 - |\mathbf{x}|^2}\, d^d x.$$

This provides a simple benchmark for Monte Carlo integration, since the exact ball volumes are known.

You may use either of the following sampling approaches:

**Method A: Sample uniformly inside the $d$-dimensional unit ball $B_d$.** Using the sampling method discussed in lecture,

$$V_{d+1} = 2V_d \left\langle \sqrt{1 - |\mathbf{x}|^2} \right\rangle_{B_d} \approx 2V_d \frac{1}{N} \sum_{\substack{i=1 \\ \mathbf{x}_i \sim \mathrm{Unif}(B_d)}}^{N} \sqrt{1 - |\mathbf{x}_i|^2}.$$

**Method B: Sample uniformly inside the cube $[-1, 1]^d$.** Define

$$g(\mathbf{x}) = \begin{cases} \sqrt{1 - |\mathbf{x}|^2}, & |\mathbf{x}| \leq 1, \\ 0, & |\mathbf{x}| > 1. \end{cases}$$

Then

$$V_{d+1} = 2 \times 2^d \langle g(\mathbf{x}) \rangle_{[-1,1]^d} \approx 2 \times 2^d \frac{1}{N} \sum_{\substack{i=1 \\ \mathbf{x}_i \sim \mathrm{Unif}([-1,1]^d)}}^{N} g(\mathbf{x}_i).$$

Method B is less efficient because points outside the unit ball make zero contribution.

**Tasks**

(a) Pick one dimension $d \geq 2$ and write a Monte Carlo code to evaluate the integral above using either Method A or Method B. Higher-dimensional calculations will be considered for extra credit.

(b) The first goal of this exercise is simply to obtain a good numerical estimate of the integral. By the end of today's in-class coding session (or today's submission deadline), submit your **best estimate** of $V_{d+1}$, together with enough numerical output to show that your code is working.

(c) Next, look more carefully at how the error decreases as the number of Monte Carlo samples $N$ increases. Make a log–log plot of the error versus $N$ and check whether the slope is approximately $-1/2$. The exact unit-ball volumes are

| $d$ | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|
| $V_d$ | $\pi$ | $\frac{4\pi}{3}$ | $\frac{\pi^2}{2}$ | $\frac{8\pi^2}{15}$ | $\frac{\pi^3}{6}$ | $\frac{16\pi^3}{105}$ | $\frac{\pi^4}{24}$ |

**What to submit.** For today's checkpoint, submit your best estimate of $V_{d+1}$ and your best preliminary figure. Both can be further refined later.